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Framework for Developing Trustworthy Industrial AI Applications

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Abstract

Artificial intelligence (AI) offers great potential to increase productivity in production engineering. Nevertheless, manufacturing companies fail to successfully establish industrial AI applications on the shopfloor. Reasons are the great complexity and intransparency of AI as well as high performance and reliability requirements of production systems. To overcome these challenges and to comply with increasing regulatory requirements (e.g., the EU AI Act), we propose a framework for guiding the development of trustworthy industrial AI applications. Based on the core requirements of the production engineering domain, we propose a risk-based approach and define four qualification dimensions: safety, robustness, transparency and adaptivity.

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1. Main text

1.1. Motivation

The recent trend towards Industry 4.0 demands the ability of smart manufacturing which includes the application of intelligent systems to enable real-time optimization for higher productivity, flexibility and sustainability [1]. A key technology of Industry 4.0 is industrial artificial intelligence (AI) [2], a methodology that focusses on the development, validation, and deployment of machine learning (ML) algorithms to solve industry-relevant problems [3]. AI aims to perform data-driven problem solving by performing functions like reasoning, self-improvement and learning which are typically associated with human intelligence [2]. Major application areas of AI in production are process optimization [4,5] quality control [6–9], and predictive maintenance [9,10].

In addition, the increasing amount of data available due to the growing use of sensor technology in production enables the use of data-driven technologies [11].

This illustrates that AI technologies offer especially for manufacturing companies the potential of optimized processes. Nevertheless, many companies are hesitant about the integration of AI technologies in their production [12]. As a result, the potential of industrial AI cannot be fully exploited.

1.2. Challenges of using industrial AI

Krauß et al. [13] and Brandstätter et al. [14] summarize the key challenges of using industrial AI by pointing out three factors influencing the use of ML application in production. The influencing factors are process and industry characteristics, data characteristics, and ML model characteristics, each of which brings its own challenges for the use of industrial AI.

The process and industry characteristics include the high confidentiality of data and information, high demands on reliability, the lack of expertise in data science and the demand for context-specific provision of understandable information. Overall, this industry is conservative and requires a high level of security (privacy, reliability, comprehensiveness). Modern AI-based technologies are often difficult to comprehend due to their lack of transparency and high degree of complexity and require investments in personnel and IT infrastructure. At the same time, there is increasing cost pressure on manufacturing companies, which makes the use of industrial AI more difficult [13,14].

The second area that complicates the implementation of industrial AI are data characteristics. A good data basis is a prerequisite for high performant ML models. However, data obtained from production processes is usually of poor quality because of insufficient data integration and data management, inhomogeneous due to multiple data sources, imbalanced, and complex with low signal-to-noise-ratios. In addition, labeling the data, which is necessary for many ML algorithms, is very time-consuming. In total, although a lot of data is recorded during the production process, without further effort it is often not available in an adequate quality required by ML models [13,14].

The third aspect which hampers the use of industrial AI are ML model characteristics. They include non-deterministic behavior, intransparent functionality, and insufficient robustness and safety. Furthermore, they are susceptible to erroneous data and data drifts. Another key challenge is the generalization of ML models from a specific use case to other tasks or processes. Overall, although ML models can generally optimize industry-relevant processes, they have properties that may limit their potential, e.g., in terms of robustness and comprehensiveness. Therefore, it is essential that design requirements for ML models are considered to avoid corresponding concerns [13,14].

1.3. Regulatory requirements on industrial AI

In addition to the previously introduced challenges – process and industry characteristics, data characteristics, and ML model characteristics – even more complexity is induced by regulatory requirements on industrial AI. Alongside the European Artificial Intelligence Act (AIA) and the Assessment List for Trustworthy AI (ALTAI), the recently published ISO/IEC 42001 will be the third dominant document for the standardization and regulation of AI systems [15]. Fulfilling these requirements for trustworthy AI becomes a great challenge for organizations given the lack of alignment and standardization between beforementioned documents. The relevant requirements are discussed in more detail in chapter 2.

1.4. Objective of this work

Design principles for the development of industrial AI applications support overcoming challenges and concerns regarding their use and meeting regulatory requirements. This is necessary to realize the potential of AI in industry.

Although design principles already exist such as guidelines for the development of AI applications, these do not usually relate to the specific requirements of the industrial sector. For example, the challenge of processing personal data plays a subordinate role in an industrial context, while the processing of large volumes of process and sensor data, must be taken into account.

In order to close this gap, this work proposes design principles for guiding development and lifecycle management of industrial AI applications in order to meet the specific requirements of the industry.

1.5. Structure of the paper

Following this introduction, in Section 2, existing design principles for industrial AI will be described and evaluated regarding their fulfilment of industrial requirements. In Section 3, we propose our framework for developing trustworthy industrial AI applications. Finally, in Section 4, the results are summarized, and an outlook is given for ensuring the implementation of the proposed design principles in industrial AI management systems.

2. Related Work

In 2019 the High-Level Expert Group on AI presented ethics guidelines for trustworthy AI including seven key requirements [16]. According to these guidelines, trustworthy AI should be lawful, ethical, and robust. The seven key requirements comprise (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, non-discrimination and fairness, (6) societal and environmental well-being, and (7) accountability. The EU AI Act proposed a draft for the first EU regulatory framework for AI in April 2021 [17]. In a risk-based approach that introduces four levels of risk including unacceptable risk, high risk, limited risk, and low and minimal risk, the requirements for AI applications depend on the individual risk level. The requirements comprise both registration and conformity obligations as well as technical aspects such as data governance, transparency, technical robustness etc. Deloitte has proposed a general framework that includes seven categories for trustworthy generative AI [18]. According to these categories, a trustworthy generative AI application must be (1) accountable, (2) robust and reliable, (3) safe and secure, (4) private, (5) transparent and explainable, (6) fair and impartial and (7) responsible. Each category contains three to four specific requirements. The National Institute of Standards and Technology (NIST) is developing an AI risk management framework [19]. The second draft from August 2022 proposes seven trustworthy characteristics including according to which AI has to be (1) valid and reliable, (2) safe, (3) fair and bias is managed, (4) secure and resilient, (5) explainable and interpretable, (6) privacy enhanced and (7) accountable and transparent.

Based on an analysis of high-level AI properties, Delseney et al. define seven major challenges of ML certification (probabilistic assessment, resilience, specifiability, data quality and representativeness, explainability, robustness, and

verifiability) and propose a simplified development process to address these challenges [20]. The German working group Plattform Lernende Systeme claims that the certification of real-world AI systems should include the certification of the algorithm, the data, the requirements specification of the application, the impact of the system on other technologies, on processes, on people, on the environment, on society, and of the system design and development tools [21]. P. Santhanam from IBM Research AI defines four specific AI-related aspects of trust that comprise (1) explainability, (2) bias and fairness, (3) robustness, and (4) transparency [22]. Based on these additional requirements for AI systems, the author proposes a holistic quality management system for ensuring AI quality along the entire AI lifecycle. Furthermore, the theoretical framework for building trustworthy AI systems in general as proposed by Li et al. comprises robustness, generalization, explainability, transparency, reproducibility, fairness, privacy preservation, and accountability as important aspects [23]. Based on this framework, the authors present a systematic approach considering the entire AI lifecycle proposing a paradigm shift from performance-driven AI to trust-driven AI.

Besides a multitude of general frameworks for AI that do not further specify the application domain, some frameworks and guidelines exist that define requirements for the use of AI applications in the industrial domain including engineering and manufacturing. For the certification of AI-supported production processes, Brandstätter et al. focus on traceability, explainability and transparency, and specify how these properties can be measured and elaborate the trade-off between performance and explainability [24]. In a Franco-German position paper, a framework for industrial and trustworthy AI is proposed to foster innovation, development, and deployment of industrial applications [25]. Focusing on the transition from Industry 4.0 to Industry 5.0, Vyhmeister et al. highlight the importance of trustworthy AI for industrial socio-technical systems [26]. For conducting a systematic literature review on trust in AI and AI-powered robotics, Emaminejad and Akhavan have defined four trust dimensions comprising explainability and interpretability, performance and robustness, reliability and safety, and privacy and security [27].

The multitude and diversity of these papers highlight the need for trustworthy AI systems, regardless of their field of application. Compared with other domains, industrial AI applications are considered particularly critical due to low quality of industrial data, economic damage potential and a conservative adoption of new technologies. In order to address the specific and individual requirements of the production domain and to promote a widespread adoption of industrial AI systems by small, medium-sized and large industrial companies, we envision the need for a trustworthy industrial AI framework that will serve as a basis for a systematic approach for industrial AI development.

3. Framework for Developing Trustworthy, Industrial AI Applications

As a framework for the development of trustworthy AI applications, we propose four core dimensions of trustworthy industrial AI: transparency, robustness, safety, and adaptivity.

Additionally, we introduce two design principles: risk-based approach and standard-compliant documentation. Representing key characteristics of trustworthy industrial AI, the core dimensions support the systematic analysis of relevant requirements to facilitate the development of trustworthy AI applications. The two design principles are intended to guide through a systematic analysis and prioritization of system requirements along the four core dimensions (see Figure 1).

3.1. Core Dimensions of Trustworthy, Industrial AI

The core dimensions correspond to four key characteristics of trustworthy industrial AI that address the domain-specific challenges of AI applications in production.

For industrial use cases, relevant requirements must be

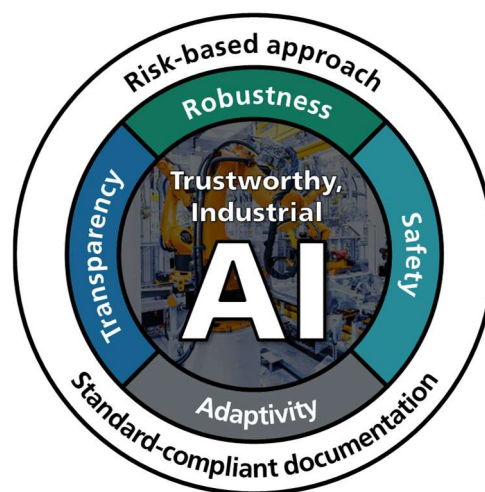


Figure 1: Visualization of the proposed framework for trustworthy, industrial AI comprising four core dimensions framed by two design principles.

identified along the core dimensions depending on the boundary conditions and regulatory requirements of individual use cases. Collected requirements shall be addressed within the development process, e.g., as part of design decision, test design, and documentation.

Dimension “Transparency”

In an industrial context, transparency of AI applications can be described as the traceability of the decision-making process from the user’s perspective. Users of AI applications in production environments typically are process experts without an in-depth knowledge of the working principles of AI. Therefore, users must be provided with insights from the decision-making process to be able to evaluate the plausibility of decisions and increase acceptance. AI applications are often perceived as black box systems, as the reasoning, rules, and input-output-functionality behind model predictions are difficult to comprehend. To enhance transparency, one option is to select AI models that inherently offer an increased degree of transparency. For instance, decision trees can be interpreted as a set of understandable IF-THEN rules. In addition, explainability techniques offer further means to increase transparency of black-box models. Examples are feature

importance methods, providing conclusions about the most important data attributes for decision making.

Dimension “Robustness”

The robustness of an AI application describes its ability to generate reproducible results within its intended range of operation where it should be robust against both data and model-related noise. Robustness is important in the context of AI-supported quality inspection, for example, to ensure a consistent high accuracy of inspection results. Reproducible and plannable AI-model behavior is also a fundamental requirement in safety-critical applications. A model-related challenge for robustness is the stochastic nature of some AI-model classes, leading to inconsistent results during model training and inference. Further dynamic influences that must be considered directly result from the production environment, leading to data noise. One example is varying lighting conditions in visual tasks. During system development, robust behavior must be ensured by representing expected variations at the operating point in the training data as well as by a suitable test design.

Dimension “Safety”

In the context of trustworthy industrial AI, safety is the ability to perform reliably facing abnormal or missing data. In contrast to robustness, the safety dimension refers to scenarios outside the intended operation range. The goal is to prevent harmful behavior, for anomalous, adversarial, or missing data inputs. While missing or corrupted data inputs can result from sensor defects, adversarial samples refer to the targeted manipulation of inputs to trigger faulty model behavior. Possible scenarios for adversarial attacks range from manipulating inputs to conceal quality defects in quality inspection tasks (economic impact) to interventions in a robot control system with potential human hazard. Trustworthy AI applications must be able to recognize such situations and react accordingly, e.g., by a system shutdown.

Dimension “Adaptivity”

Adaptivity describes an AI application’s ability to recognize changes in the operating environment over time and adapt accordingly to ensure long-lasting system functionality. In industrial environments, AI applications usually are subject to changing boundary conditions over time, resulting in a decreasing accuracy of the system. Dynamic influences on industrial AI applications are manifold. Changes in the operating conditions can result from products (e.g., new product variants), machines and equipment (e.g., change process parameters or tool wear), and the physical environment (e.g., changing climate conditions). In general, two categories of dynamics, so-called data drifts, can be distinguished in the context of AI applications [28]:

- **Covariance drift:** The distribution of input data changes compared to the data used for training, e.g., resulting from changed settings for data acquisition.

- **Concept drift:** The relation between input data and the target variable changes compared to the learned relation in training, e.g. if new product variants with new characteristics are introduced.

To ensure long-lasting functionality, an AI application must be capable of detecting drifts and adapting accordingly. A common measure for adapting to concept or data drifts is retraining a model with new data that reflects the changed context.

3.2. Design Principles for Trustworthy, Industrial AI

The two design principles risk-based approach and standard-compliant documentation shall guide the analysis and prioritization of system requirements in the four core dimensions.

Within the risk-based approach, relevant system risks are identified, assessed, and system requirements for risk mitigation are derived. In risk assessment, identified risks are evaluated in terms of probability and damage potential in order to derive a risk prioritization. In the context of industrial AI applications, the damage potential of system malfunctions ranges from economic losses, e.g., when quality defects are overlooked during inspection, to personal hazard caused by misguided controlling of industrial robots. Results of the risk-based approach are transparency of the risks of an AI application and a catalogue of system requirements intended for risk mitigation.

The second objective of the design principles is to comply with industry-specific documentation standards and regulations. In the medical sector, for example, regulations require an increased level of technical documentation and specification, which also applies to AI applications. These requirements contrast with some inherent characteristics of AI. Since AI models learn their behavior from experience (data) and are not programmed purely deterministically, not all system properties can be specified completely. In addition, some AI model classes have a stochastic nature and do not behave fully deterministically. Accordingly, it must be ensured that a sufficient degree of system specification is achieved by means of an appropriate test design and coverage during system development.

4. Conclusion and Outlook

Industrial AI applications face challenges in process and industry characteristics as well as data characteristics and ML model characteristics. To address these, a framework for the development of trustworthy industrial AI, consisting of four core dimensions and two design principles, is proposed.

The first core dimension, **transparency**, focusses on increasing the traceability and understandability of the decision-making process of AI applications to provide insights into the reasoning behind the AI model's predictions. Transparency can be enhanced by selecting AI models with inherent transparency and using explainability techniques.

The second core dimension, **robustness**, emphasizes the ability of AI applications to produce consistent and

reproducible results within the intended range of operation. It involves addressing the stochastic nature of some AI models and accounting for inconsistent results during model training due to data noise.

The third core dimension, **safety**, focuses on the reliable performance of AI applications when dealing with abnormal or missing data inputs. It aims to prevent harmful behavior caused by anomalous, adversarial, or missing data. Trustworthy AI applications should be able to recognize and react to such situations, including adversarial attacks.

The fourth core dimension, **adaptivity**, refers to an AI application's ability to recognize and adapt to dynamic changes in the operating environment. Industrial AI applications often face dynamic boundary conditions that can affect system accuracy over time. Adaptivity involves detecting and responding to data by retraining the model with new data that reflects the changed environment.

The two design principles, **risk-based approach**, and **standard-compliant documentation**, support the systematic analysis and prioritization of system requirements in the four core dimensions. The risk-based approach involves identifying and assessing relevant system risks, prioritizing them based on probability and potential damage, and developing measures for risk mitigation. Standard-compliant documentation ensures that industry-specific standards and regulations are met, balancing the inherent characteristics of AI models with the need for system specification.

Overall, this framework aims to serve as a basis for a systematic approach to requirement engineering and the development of trustworthy industrial AI applications. By addressing the introduced dimensions and principles, developers can prioritize requirements, mitigate risks, and ensure long-lasting functionality and reliability in industrial AI applications.

The proposed framework builds the basis for an organization's AI (quality) management system, which represents a systematic approach to ensure the compliance with the dimensions and design principles for trustworthy industrial AI during the development and operation of AI applications. Based on this framework, our further research will focus on the requirement-conformant design and implementation of such management systems aiming to establish AI guidelines in an organization to finally drive the widespread adoption of industrial AI applications.

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